

200G LPO: Design Challenges and Latest Test Data

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Abstract: Linear Pluggable Optics (LPO) is a promising technology to reduce power consumption at 200 Gbps. This paper will review the benefits, requirements and challenges of implementing LPO technology at 200 Gbps. © 2026 The Author(s)

1. Introduction

The requirement for high bandwidth interconnects operating at 200G per lane is ramping to high volume in data centers to support the demand driven by AI workloads. Optical interconnects are required for the scale-out of AI hardware beyond the rack into larger compute clusters, and there is also interest in using low-power optical interconnects for scale-up applications to increase both reach and bandwidth within the rack.

Both cost and power of optical interfaces have increased as the data rates have increased. For next-generation systems, the power consumption for optics can exceed multiple kilowatts per rack, and optical transceiver cost can be a significant fraction of total network hardware cost. Power consumption associated with optical interfaces is becoming an issue at 200G/lane, where it can exceed 25 W for a single 1.6 Tbps optical transceiver. This introduces significant thermal and cooling design challenges for AI hardware racks, which are already often operating at the limit of available data center cooling and power feed capability.

Linear Pluggable Optics (LPO) technology offers a low-power alternative to retimed optical modules either in the traditional pluggable module format or in the emerging Near Package Optics (NPO) format and Co-Packaged Optics (CPO) formats.

2. Linear Pluggable Optics (LPO) Overview

An LPO link is shown in Figure 1. LPO links are linear and do not contain retimers or digital signal processors (DSP) to retime the signal in the module along the link. Any retiming is performed in the host transmitter port or host receiver port. This saves considerable power in the link.

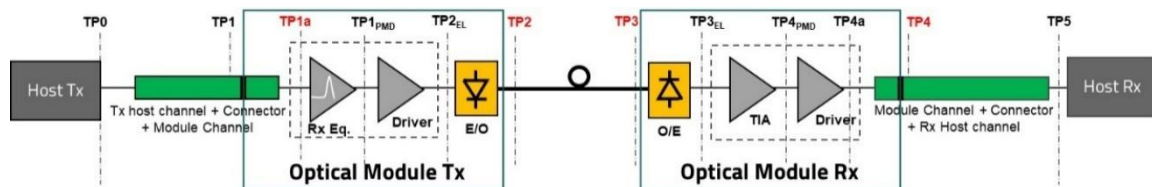


Figure 1. Block diagram of an LPO link.

The transmitter portion of the module consists of three key components: an input equalizer, an automatic gain control (AGC) and a modulator. The input equalizer supports equalization of the transmitter host port in conjunction with the host transmitter port finite impulse response (FIR) equalizer. The AGC maintains consistent signal amplitude to the modulator, ensuring the extinction ratio (ER) and optical modulation amplitude (OMA) does not vary with changes on the electrical input. The modulator converts the electrical signal to an optical signal.

The receiver portion consists of a photodiode and transimpedance amplifier (TIA) to convert the optical signal to an electrical signal. The TIA requires an AGC to provide constant output swing as the optical input power varies. Additional diagnostic features can be included in both the transmitter and receiver to support link health monitoring and link failure analysis.

At 200G/lane, a 1.6T LPO module is expected (from internal simulation of TIAs and Drivers and realistic estimates of laser power and ancillary module function power) to consume less than 10 W. This compares very favorably

against a fully retimed module which currently consumes up to 25 W using 3nm DSP technology and is expected to reduce to approximately 20 W with 2nm DSP technology.

3. LPO Methodology and Interoperability

Interoperability is a key requirement for any technology to be deployed at scale in a data center. The LPO methodology has been developed to enable “plug-and-play” interoperability without having to co-optimize the port and module together. The key to this is full equalization of the host transmitter electrical port using the host port transmitter FIR and the input continuous time linear equalizer (CTLE) of the driver integrated circuit (IC). This combination gives a consistent eye shape with different port losses — the eye amplitude varies with electrical port loss and equalization but this is accounted for by the driver AGC function — and additionally has the benefit of maximizing the signal-to-noise ratio (SNR). Maximizing the SNR (without over equalization or peaking) at the transmitter enables best bit error rate (BER) floor performance from an SNR perspective. It should be noted that the port equalization does not require high precision and achieving full equalization within approximately ± 1 dB is adequate.

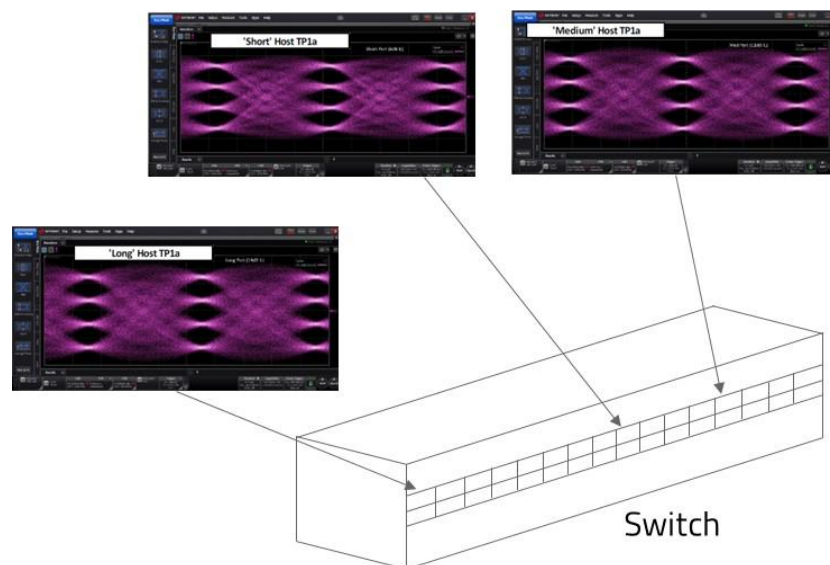


Figure 2. Measured eye diagrams of different port loss when fully equalized.

The signal in an LPO link is fully recovered at the host port receiver and the LPO link contains no retimers or DSPs between the transmitter and receiver serializer/deserializer (SerDes). It is also important that the electrical host ports have good return loss to minimize reflections as reflections can be considered as non-linear effects.

The required capabilities of the receiver are less than the typical implementations found in widely deployed merchant SerDes designs. The Optical Internetworking Forum (OIF) and Linear Pluggable Optics Multi-Source Agreement (LPO-MSA) standards define an equalizer with a capability of at least 3 precursors and 18-post cursors for 100 Gbps. The receiver for 200G/lane is expected to specify a similar type of equalizer.

4. 200G/lane LPO Considerations

LPO modules using 100G/lane signaling showed very good performance with BER floors of 10^{-12} and 10^{-9} in initial production deployment across a 64-port switch with modules from different vendors [1], and can likely exceed fully retimed module performance if links are fully optimized. Extending the line rate to 200G/lane and achieving the same performance requires the implementation and technology to improve as is the case for other 200G/lane applications defined by the OIF and Institute of Electrical and Electronics Engineers (IEEE) standards. The existing LPO methodology can still be applied if the technology can be optimized to give adequate performance for the electrical channel in the host ports. Simulation indicates that electrical host port losses up to 26 dB die-to-die can be equalized if care is taken to provide equalization at the correct points along the link to maintain adequate SNR.

Simulation indicates that maintaining a host port return loss of better than 10 dB is required to maintain the BER performance. Early measurements of host electrical ports indicate return losses of better than 10 dB across all ports are achievable, which aligns with the simulations and shows that host return loss is not a dominant factor below that threshold.

Bandwidth is another important consideration for link performance. This is true for all types of systems (not only for LPO systems), however as LPO links are unretimed and the impairments can build up, bandwidth can be more important in LPO links. Module transmitter bandwidth in particular is important as it reduces the equalization requirements and can relax the linearity requirements on the components. At 100G/lane, the transmitter bandwidth was generally above Nyquist (~ 26 GHz) and getting up to 35 GHz of bandwidth was not difficult. At 200G/lane, achieving the same equivalent bandwidth (53 GHz for Nyquist and up to 70 GHz improved performance) is possible but not as easy as it was at 100G/lane. Figure 3 shows how reflections and transmitter bandwidth affect the link BER performance. Further details about these simulations can be found in [2] and [3].

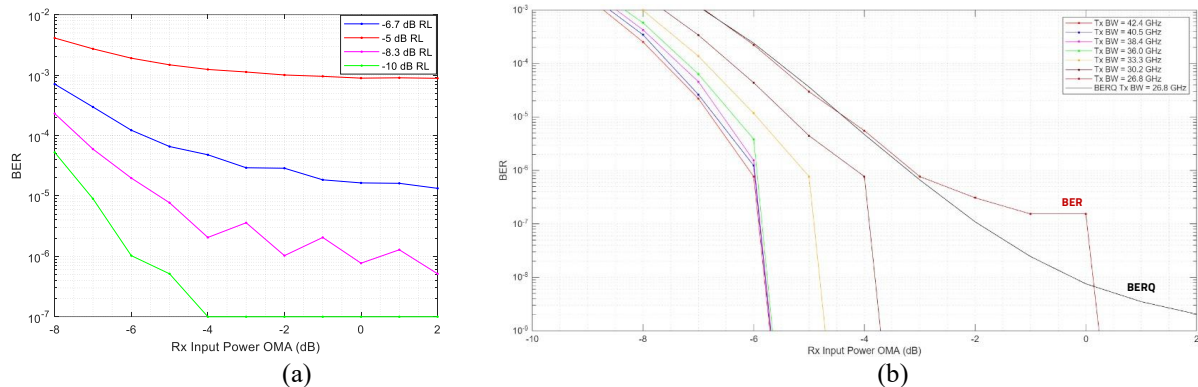


Figure 3. (a) Simulated BER performance of an LPO link with an electrical port loss of 13 dB at Nyquist. The return loss is artificially varied. (b) Effect of transmitter bandwidth on BER link performance (simulated).

5. 200G/lane LPO Performance

Extensive system simulations of 200G/lane LPO systems have been made and indicate that good performance can be achieved provided the reflections in the electrical ports can be controlled and the transmitter bandwidth is maintained. The block diagram of the system simulation is shown in Figure 4, and the simulated BER performance is shown in Figure 5. The host transmitter equalization was set up as detailed in Section 3 (i.e. the host transmitter FIR and driver CTLE fully equalizes the electrical port). The simulations include electrical crosstalk, laser relative intensity noise (RIN), TIA noise, and modulator nonlinearity. The host ports were simulated with 10 dB return loss. While the frequency responses are synthesized in these simulations, the key impairments are close to the maximum values expected. The simulations show that reasonable BER performance can be expected across the range of port losses up to 26 dB. Reducing the impairments, particularly return loss, will improve the BER performance. Further details about these simulations can be found in [4].

Another aspect that the simulations show is that with higher port loss, adding equalization before the analog-to-digital converter (ADC) is beneficial in reducing the BER floor. The equalization before the ADC helps to reduce the impact of ADC quantization noise in the system. This is most clearly seen in the simulation with 26 dB electrical port loss where the BER floor can be reduced to less than 10^{-6} from 10^{-3} .

The receiver equalizer capability could also be increased by adding more complex equalization techniques such as maximum likelihood sequence estimation (MLSE). Such equalization techniques have been tested in other applications and generally show SNR improvements of 0.5 dB to 1.5 dB. This could improve the error floor by one to two decades.

The first 200G/lane switches were announced mid-2025, with early versions of production switches becoming available in late 2025. As a result, there have been limited opportunities to make LPO measurements in a switch environment. Creating representative application channels using module compliance boards (MCBs) has proven difficult due to reflections that degrade performance, however Kang [1] achieved BER floors of 10^{-9} using an MCB approach.

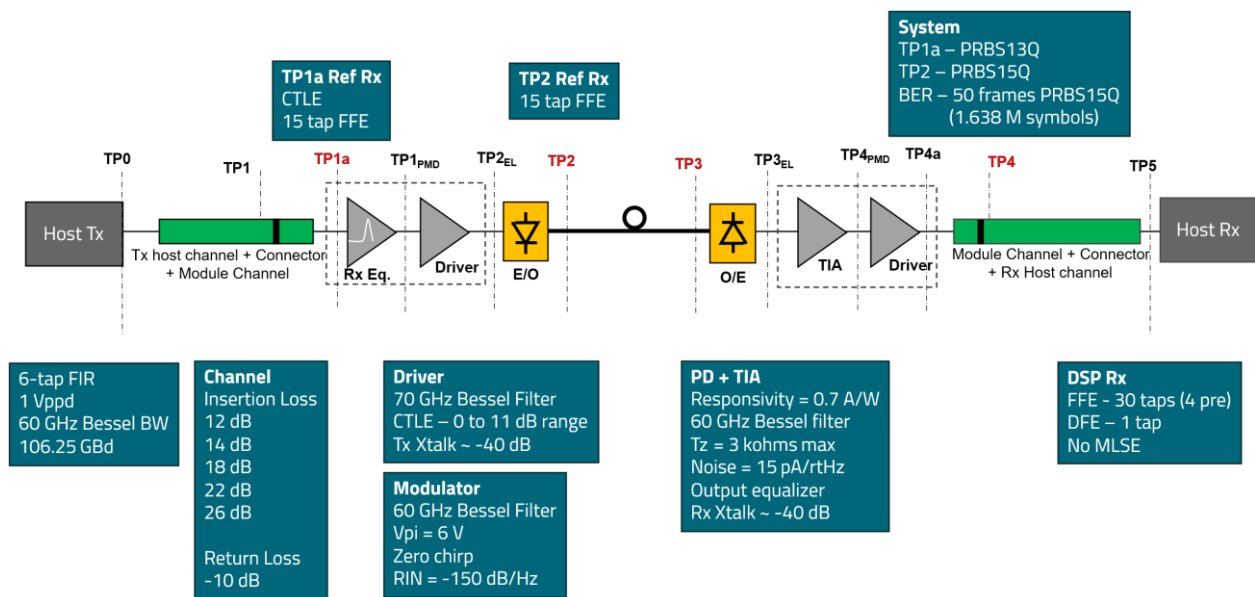


Figure 4. Block diagram of 200G/lane system simulation.

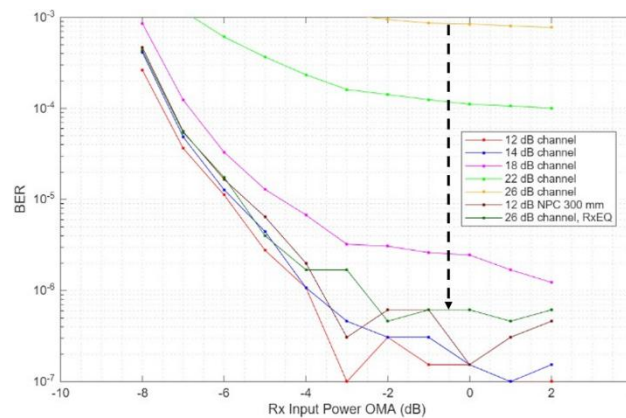


Figure 5. Simulated BER performance of 200G/lane link with various port losses.

6. Summary

LPO-based systems operating at 100G/lane have shown good performance in both laboratory and field trial scenarios which gives confidence that LPO is a viable solution for large-scale deployment. Extending the performance to 200G/lane seems reasonable provided electrical reflections in the host ports are controlled and transmitter bandwidth is maintained at a reasonable value. The recent advent of 200G/lane switches should enable further progress and improvement in LPO performance.

7. References

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- [2] Ed Frlan, Mark Kimber, Bill Kirkland, "224G Linear updated channel definition and additional TP1a/TP2 simulations," OIF presentation, oif2025.034.01, Feb 2025
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